

Ans Key Quiz 3
HS 239, Winter 2026¹

1. This question relates to Marshallian inefficiency of sharecropping, as discussed in class. Assume that an agricultural production process has production function $y = \sqrt{e}$ and cost as $c(e) = ke$, where e is the intensity of effort.

a) If the landlord tills his/her own land, what must be (i) the level of effort and (ii) landlord's income?

b) If the landlord hires a tenant, and opts for a fixed rent R (not so high that tenant's profit is zero or negative), what must be the (i) tenant's income (net of effort cost), (ii) landlord's income?

c) Now suppose the landlord demands $s\%$ of output, leaving $(1 - s) \%$ to the tenant. Re-do part (b).

d) Now assume that the landlord fixes R to be what he was getting in part (c) and offers a fixed rent contract to tenant ($s = 0$). Re-do part b.

e) Argue why (d) is superior to (c) ($3+3+4+3+5=18$)

Ans Key

This relates to page 1-2 of topic 3B Note and an example done in class.²

a) *Landlords income net of effort cost is*

$$\sqrt{e} - ke$$

¹**Statutory Caveat:** The key may not be error free. You are requested to go through the solutions and let me know if there are any discrepancies. There may be more than one *logical* way to answer a question.

²So those who attended the lecture, I am sure that they will score full marks. Those who did not attend the lecture, I expect them to follow regularly. Hence, I am sure that the latter group will also get full marks.

Maximizing this with respect to e

$$\begin{aligned}\frac{1}{2\sqrt{e}} &= k \\ \rightarrow e_{FB} &= \frac{1}{4k^2}\end{aligned}$$

Landlord's income net of effort is

$$\begin{aligned}&\sqrt{e} - k * e \\ &= \frac{1}{2k} - \frac{1}{4k} \\ &= \frac{1}{4k}\end{aligned}$$

b) Now the tenant supplies the effort. His/her income net of effort cost is

$$\sqrt{e} - ke - R$$

Maximizing with respect to e , we get

$$e = \frac{1}{4k^2} = e_{FB}$$

Then, income of tenant net of effort cost

$$\begin{aligned}&\sqrt{e} - k * e - R \\ &= \frac{1}{2k} - \frac{1}{4k} - R \\ &\quad \frac{1}{4k} - R\end{aligned}$$

c) In this part there is no R and the contract is pure sharecropping. Here, the tenant maximizes

$$(1 - s)\sqrt{e} - ke$$

The FOC is

$$\begin{aligned}(1 - s) * \frac{1}{2\sqrt{e}} &= k \\ \rightarrow e &= \frac{1}{4k^2} (1 - s)^2\end{aligned}$$

Tenant's income net of effort cost

$$\begin{aligned}
& (1-s) \sqrt{\frac{1}{4k^2} (1-s)^2} - k \frac{1}{4k^2} (1-s)^2 \\
= & \frac{(1-s^2)}{2k} - \frac{(1-s)^2}{4k} \\
= & \frac{(1-s)^2}{4k}
\end{aligned}$$

The landlord's income is

$$\begin{aligned}
& s\sqrt{e} \\
= & \frac{s(1-s)}{2k}
\end{aligned}$$

d) Suppose the landlord now fixes $R = \frac{s(1-s)}{2k}$, where s is the particular share value (e.g., $s = s_0$). Then tenant's income net of effort cost is

$$\begin{aligned}
& \sqrt{e} - ke - R \\
= & \sqrt{e} - ke - \frac{s_0(1-s_0)}{2k}
\end{aligned}$$

Maximizing with respect to e ,

$$e = \frac{1}{4k^2}$$

Then tenant's income net of effort cost now

$$\begin{aligned}
& \frac{1}{2k} - \frac{1}{4k} - \frac{s_0(1-s_0)}{2k} \\
= & \frac{1}{4k} - \frac{s_0(1-s_0)}{2k}
\end{aligned}$$

e) *Tenant's income net of effort cost (under sharecropping) is*

$$\frac{(1-s_0)^2}{4k}$$

Then, we have, the difference between the two situations for tenant

$$\begin{aligned}
& \left[\frac{1}{4k} - \frac{s_0(1-s_0)}{2k} \right] - \frac{(1-s_0)^2}{4k} \\
&= \frac{1}{4k} - \frac{2s_0(1-s_0) + (1-s_0)^2}{4k} \\
&= \frac{1}{4k} - \frac{1-s_0^2}{4k} \\
&= \frac{s_0^2}{4k} > 0
\end{aligned}$$

Thus the tenant strictly prefers situation (d) over (c). The landlord is indifferent between (d) and (c). Moving to d from c at least makes one party better off without making none worse off. Therefore d is superior to c.³

Part 2 (MCQ)

a) In a model of sharecropping, higher disturbance in production will create higher share for tenant (True/false)

False, as the share does not depend on disturbance term (σ^2).

b) The utility function $u = \sqrt{x}$ would exhibit precautionary savings. (True/ False)

True, as $u''' = \frac{3}{8x^{\frac{5}{2}}} > 0$. Thus, u' is convex.

c) The utility function , $u = \alpha x - \frac{\beta}{2}x^2$, $\alpha, \beta > 0$, would exhibit precautionary savings.(True/false)

False, as $u''' = 0$.

d) In a model of sharecropping, higher disturbance in production will create higher fixed wage or lower rent for tenant. (True/false) .

True. Assume the contract to the tenant is $sY - R$, $R > 0$ implying rent. In equilibrium, $\bar{U} = se - R - \frac{\beta_T}{2}s^2\sigma^2 - ce^2$, where $\bar{U}$ is the reservation

³Of you want to show off your wisdom, you may say d is Pareto superior to c.

utility. If everything is optimally set up, higher σ^2 means R to fall (becomes a lower positive value or higher negative value)

e) Higher tax rates would induce more tax compliance if the utility function is CARA/DARA/IARA. (Tick)

DARA

f) In the model of tax evasion with a constant moral cost of hiding income, the proportion of people under-declaring their income will go up with higher income. (T/F)

*False. Suppose the moral cost is ϕ . Then the agent does not evade if $\phi > k_0 = t(1 - p - Fp) * u'((1 - t)Y)$. We have $\frac{dk_0}{dY} = t(1 - p - Fp) * (1 - t) * u''((1 - t)Y) < 0$. Therefore, the percentage of people who declare their income goes up, under-declaration goes down.*

g) A gentleman has CRRA utility function. His income is Y and under declaration is $U = Y - X$. The ratio U/Y , with increasing income, will (a) stay constant, (b) increase, (c) decrease. (Tick)

With CRRA utility function, the ratio X/Y is constant. Here, $\frac{U}{Y} = 1 - \frac{X}{Y}$. As $\frac{X}{Y}$ is constant for all income, U/Y is also constant for all income